Madhav Tripathi

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Education

University of Illinois Urbana-Champaign, Computer Science (B.S.) — Urbana-Champaign, IL, Aug 2023 – May 2028

- GPA: 3.70/4.00; James Scholar (2024–present); Dean's List (Fall 2024).
- Coursework: Data Structures; Linear Algebra; Computer Architecture.
- In progress: Probability & Statistics; Systems Programming; Database Systems; Computer System Organization.

Experience

Siebel School of Computing and Data Science, Research Assistant, Shepherd Research Lab — Urbana, IL, Mar 2026 – present, 1 month

Research assistant in Shepherd Lab, developing methods to improve context coherence and compute on masked/discrete diffusion-based language models.

- Designed and implemented a reproducible ablation pipeline for a training-free diffusion-based rendering, benchmarking MVSplat against four SEVA variants with Hydra-configured experiment orchestration and CLIP-guided schedule modulation.
- Isolated per-scene and per-mode artifacts and used metrics such as PSNR, SSIM, LPIPS, and MUSIQ to quantify diffusion work and quality under different conditions.

Siebel School of Computing and Data Science, Research Assistant, ScribeAR — Urbana, IL, Feb 2026 – present, 2 months

Research assistant on ScribeAR, building accessible real-time captioning and transcription systems.

- Currently developing a general-purpose Redux state migrator to support the addition of new UI components without custom migrations.
- Designing and implementing a speaker-aware transcription architecture that integrates with a server-side ML diarization service over WebSockets, enabling real-time speaker-change detection, shared typed event contracts, and live frontend speaker state updates.
- Implemented and integrated a new StreamText transcription provider end-to-end across frontend state management, provider runtime logic, configuration UI, persistence migration, and provider registry wiring in a TypeScript monorepo with a React/Vite + Redux web client, Fastify-based backend, shared schema/library packages, and PostgreSQL infrastructure.

Siebel School of Computing and Data Science, Course Assistant, Introduction to Computer Science II — Urbana, IL, Aug 2025 – present, 8 months

Course assistant for CS II, supporting students in C++ programming and core data structures.

- Led in-person and online help sessions; debugged C++ code and reinforced core concepts with structured walkthroughs.
- Facilitated discussion sections focused on hands-on implementation of course concepts.
- Built PrairieLearn practice activities for struggling students and led weekly extra-practice sessions.

Siebel School of Computing and Data Science, Course Assistant, Introduction to Computer Science I — Urbana, IL, Jan 2025 – present, 1 year 3 months

Course assistant for CS I, helping students build strong foundations in Java and introductory CS.

- Held office hours; guided Java fundamentals, MP debugging, and quiz preparation.
- Supported a ~200-student discussion section and mentored ~30 students on a semester-long machine project.

Projects

ASL Transcription Service (CLIP + Django REST) UIUC

Real-time ASL alphabet transcription service using webcam capture, a Django REST API, and a CLIP-based classifier. Aug 2025 – Jan 2026

- Built a real-time ASL alphabet classifier (24 classes; J/Z excluded) with webcam capture → Django REST → top-1 prediction + confidence.
- Fine-tuned CLIP ViT-B/32 via prompt-tuned text tokens while optimizing the vision encoder; exported the best checkpoint for deployment.
- Implemented end-to-end train/infer/eval with checkpointing, cached inference, CI, and metrics dashboards (confusion, top-k, calibration/ECE); achieved 98.46% best validation accuracy.

Viral Protein Mutation Modeling Birla Vidya Niketan / Ritsumeikan University

Mentored research project on viral protein structure/function and mutation pathways, presented at conferences/fairs. Jan 2022 – Nov 2023

- Developed and presented a mentored project (guided by the high school CS department head) on viral protein structure/function and mutation pathways.
- Organized a dataset pipeline from protein databank files and prototyped a CNN-based predictor on an engineered representation of the spike protein; documented methodology, limitations, and findings.

Skills

Languages

Engineering Tools

Data & Vision

Machine Learning